PRECOMPOSITION ON CONTINUOUS ALTERNATING MAPS IS NOT ANALYTIC IN CHARACTERISTIC 2

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ABSTRACT. For normed spaces E, E', F over a nontrivially normed field k and a finite set ι, precomposition $f \mapsto (m \mapsto m \circ (f, \dots, f))$ sends continuous linear maps $E \to E'$ to continuous linear maps between spaces of continuous alternating maps. It is C^n for every $n \in \mathbb{N} \cup \{\infty\}$ in every characteristic [1], and analytic when $|\iota|!$ is invertible in k (by polarisation). We show that it is *not* analytic in general: over a complete ultrametric field of characteristic 2 with infinite residue field that is not spherically complete, with $E = E' = \ell^\infty$ and $|\iota| = 2$, analyticity fails at 0. The proof is a finite parity identity for determinants in characteristic 2, glued along ℓ^∞. The result is formalized in Lean 4/Mathlib in the form "the general C^ω statement implies `False`".

1. STATEMENT

Notation. For normed spaces, $\mathcal{L}(E, E')$ is the space of continuous linear maps with the operator norm, and $\mathrm{Alt}^\iota(E, F)$ is the normed space of continuous alternating ι-linear maps $E^\iota \to F$, i.e. those vanishing whenever two arguments coincide; we write Alt^2 when $|\iota| = 2$. For a polynomial f and a monomial μ, $\mathrm{coeff}_\mu f$ is the coefficient of μ in f. We write $c_0 \subset \ell^\infty$ for the null sequences, $\mathbf{1}_S$ for the indicator of a set S, and $\mathbf{1} = \mathbf{1}_{\mathbb{N}}$.

Throughout, k is a complete non-archimedean valued field such that

(K1) $\mathrm{char}\, k = 2$;
(K2) the residue field of k is infinite;
(K3) there is a sequence of closed balls $B(c_n, r_n)$ with $|c_{n+1} - c_n| \le r_n$, $r_0 > r_1 > \cdots$ and $\bigcap_n B(c_n, r_n) = \emptyset$; in particular k is not spherically complete.

Such fields exist; an explicit one is given in §6. Let $E = \ell^\infty = \ell^\infty(\mathbb{N}, k)$ with the sup norm, a Banach algebra under pointwise multiplication; write $m_a \in \mathcal{L}(E, E)$ for multiplication by $a \in E$, so that $\|m_a\| \le \|a\|$, and e_i for the unit vectors. For $x, y \in E$ let

$$W(x, y) := x \otimes y + y \otimes x \in \ell^\infty(\mathbb{N} \times \mathbb{N}, k), \qquad W(x, y)_{pq} = x_p y_q + x_q y_p,$$

and let $F \subseteq \ell^\infty(\mathbb{N} \times \mathbb{N}, k)$ be the closed linear span of all $W(x, y)$. Every element of F is a symmetric matrix with zero diagonal (as $W(x, y)_{pp} = 2x_p y_p = 0$) and a uniform limit of matrices of finite rank (as $W(x, y)$ has rank ≤ 2). By the ultrametric inequality $\|W(x, y)\| \le \|x\|\|y\|$, and $W(x, x) = 2\, x \otimes x = 0$; so $W \colon E \times E \to F$ is a bounded alternating bilinear map of norm ≤ 1. Let

$$Q \colon \mathcal{L}(E, E) \to \mathcal{L}\big(\mathrm{Alt}^2(E, F), \mathrm{Alt}^2(E, F)\big), \qquad Q(f)(m) = m \circ (f, f).$$

Theorem 1. *Q is not analytic at 0. In particular Q is not C^ω.*

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The counterexample, its proof and the Lean formalization described here were produced with Anthropic's Claude (in Claude Code). Theorems 1 and 2 are machine-checked in Lean 4 with Mathlib; see §8. The informal text has not yet been reviewed by a human expert.

Theorem 2. *The statement "for all nontrivially normed fields k, all normed spaces E, E', F over k and all finite ι, precomposition $\mathcal{L}(E,E') \to \mathcal{L}(\mathrm{Alt}^\iota(E',F), \mathrm{Alt}^\iota(E,F))$ is C^ω" is false. So is the weaker statement in which k ranges only over complete ultrametric fields of characteristic 2, $E = E'$ and F range only over Banach spaces, and $|\iota| = 2$.*

Theorem 2 follows from Theorem 1 and the existence of a field with (K1)–(K3). The rest of this note proves Theorem 1 in four steps.

2. Step 1: analyticity forces a bilinear lift

Lemma 3. *Let $Q\colon X \to Z$ be a map between normed spaces with $Q(tf) = t^2 Q(f)$. If Q is analytic at 0 then there is a bounded bilinear $P\colon X \times X \to Z$ with $P(f,f) = Q(f)$ for all f.*

Proof. Let $\sum_n p_n$ be a power series of Q at 0. For fixed f the one-variable power series $\sum_n t^n\, p_n(f,\dots,f)$ and the polynomial $t^2 Q(f)$ agree for small t. Power series coefficients in one variable are unique over any nontrivially normed field, so $p_2(f,f) = Q(f)$. Take $P = p_2$. (No division by 2 occurs, and P need not be symmetric.) □

So assume that a bounded bilinear P with $P(f,f) = Q(f)$ exists, and put, for $a, b, x, y \in E$,

$$\Phi(a,b;x,y) := P(m_a, m_b)(W)(x,y) \in F.$$

Then Φ is 4-linear, and since $\|m_a\| \le \|a\|$ and $\|W\| \le 1$ we have

$$\|\Phi(a,b;x,y)\| \le C\|a\|\,\|b\|\,\|x\|\,\|y\|, \qquad C := \max(1, \|P\|).$$

Moreover

$$\Phi(a,b;x,x) = 0, \qquad \Phi(a,a;x,y) = W(ax,ay). \tag{1}$$

Since $2 = 0$, the first identity gives the symmetry $\Phi(a,b;x,y) = \Phi(a,b;y,x)$. We will contradict the existence of such a Φ.

3. Step 2: a finite parity identity

Fix a finite block $I \subset \mathbb{N}$ of even size $2m$ and $a', b' \in E$ vanishing on I with $\|a'\|, \|b'\| \le 1$. Let X_i, Y_i $(i \in I)$ be indeterminates. Expanding Φ multilinearly in $x = \sum_{i\in I} X_i e_i$ and $y = \sum_{i\in I} Y_i e_i$ defines polynomials

$$G_{pq} := \Phi(a' + x,\, b' + y;\, a' + x,\, b' + y)_{pq} \in k[X,Y] \qquad (p,q \in I),$$

whose values at points $x, y \in k^I$ (vectors supported on I) are the entries of an $I \times I$ matrix $G(x,y)$. The matrix G is symmetric with zero diagonal, each G_{pq} has degree ≤ 2 in X and ≤ 2 in Y, its part of bidegree $(2,2)$ is $g_{pq} := \Phi(x,y;x,y)_{pq}$, and $G_{pq} - g_{pq}$ has total degree ≤ 3.

Lemma 4. *For $p \ne q$ we have, in $k[X,Y]$,*

$$g_{pq}(Y,X) = g_{pq}(X,Y) + X_p^2 Y_q^2 + X_q^2 Y_p^2 \qquad \text{and} \qquad g_{pq}(X,X) = 0.$$

Proof. As k is infinite it suffices to check these at points $x, y \in k^I$. By the symmetry in the last two slots, $g(y,x) = \Phi(y,x;y,x) = \Phi(y,x;x,y)$, so by bilinearity in the first two slots and (1)

$$\begin{aligned} g(x,y) + g(y,x) &= \Phi(x+y, x+y; x, y) + \Phi(x,x;x,y) + \Phi(y,y;x,y) \\ &= W\big((x+y)x, (x+y)y\big) + W(x^2, xy) + W(xy, y^2). \end{aligned}$$

Expanding, $W((x+y)x, (x+y)y) = W(x^2,xy) + W(x^2,y^2) + W(xy,xy) + W(xy,y^2)$. The first and last terms cancel against the two other summands and $W(xy,xy) = 0$, which leaves $W(x^2,y^2)$, whose (p,q) entry is $x_p^2 y_q^2 + x_q^2 y_p^2$. Finally $g(x,x) = W(x^2,x^2) = 0$. □

Let τ be the automorphism of $k[X,Y]$ swapping $X_i \leftrightarrow Y_i$, and δ the substitution $Y_i := X_i$.

Lemma 5 (balanced coefficients)**.** *Let $f \in k[X,Y]$ with $\tau f = f$ and $\operatorname{char} k = 2$. For every $S \subseteq I$ the coefficient of $\prod_{i\in S} X_iY_i$ in f equals the coefficient of $\prod_{i\in S} X_i^2$ in δf.*

Proof. The coefficient of $\prod_S X_i^2$ in δf is the sum of the coefficients in f of all monomials $X^\alpha Y^\beta$ with $\alpha+\beta = 2\cdot\mathbf{1}_S$. The involution $(\alpha,\beta)\mapsto(\beta,\alpha)$ of this set of monomials preserves coefficients (as $\tau f = f$), and its only fixed point is $\alpha=\beta=\mathbf{1}_S$; all other terms cancel in pairs. □

Proposition 6. *Let M be a perfect matching of I and $\nu := \prod_{i\in I} X_iY_i$. Then*

$$\operatorname{coeff}_\nu \prod_{\{p,q\}\in M} G_{pq} = 1.$$

Proof. The monomial ν has total degree $4m$, the maximum possible, so only the top parts contribute:

$$\operatorname{coeff}_\nu \prod_M G_{pq} = \operatorname{coeff}_\nu \prod_M g_{pq}.$$

Put $g'_{pq} := g_{pq} + X_p^2Y_q^2$. By Lemma 4,

$$\tau g'_{pq} = g'_{pq} \qquad \text{and} \qquad \delta g'_{pq} = X_p^2X_q^2.$$

A product containing a factor $X_p^2Y_q^2$ has no squarefree monomial, so replacing every g_{pq} by g'_{pq} does not change the coefficient of ν. By Lemma 5, applied to the τ-invariant product $\prod_M g'_{pq}$, this coefficient is the coefficient of $\prod_{i\in I} X_i^2$ in

$$\delta \prod_M g'_{pq} = \prod_{\{p,q\}\in M} X_p^2X_q^2 = \prod_{i\in I} X_i^2,$$

namely 1. □

Lemma 7. *Let A be a symmetric matrix with zero diagonal over a commutative ring of characteristic 2, of even size $2m$. Then $\det A = \sum_M \prod_{\{p,q\}\in M} A_{pq}^2$, the sum being over perfect matchings; and the number of perfect matchings, $(2m-1)!!$, is odd.*

Proof. In $\det A = \sum_\sigma \prod_i A_{i\sigma(i)}$ (all signs are 1), σ and σ^{-1} contribute equally by symmetry, so only involutions survive; those with a fixed point contribute 0; a fixed-point-free involution is a perfect matching and contributes $\prod A_{pq}^2$. □

Corollary 8. $\operatorname{coeff}_{\nu^2} \det G = 1$. *Consequently, by* (K2)*, there are $x,y \in k^I$ with $|x_i|,|y_i| \le 1$ and $|\det G(x,y)| \ge 1$.*

Proof. Squaring is additive on coefficients in characteristic 2, so by Lemma 7 and Proposition 6

$$\operatorname{coeff}_{\nu^2} \det G = \sum_M \Big(\operatorname{coeff}_\nu \prod_{\{p,q\}\in M} G_{pq}\Big)^2 = (2m-1)!!\cdot 1 = 1,$$

because $(2m-1)!!$ is odd. A polynomial with a coefficient of absolute value ≥ 1 has Gauss norm ≥ 1; normalising by its largest coefficient and reducing modulo the maximal ideal gives a nonzero polynomial over the infinite residue field, which has a non-vanishing point; lift it to the unit polydisc. □

The point is that the coefficient 1 does not depend on the unknown lift P: all of its freedom is carried by τ-invariant terms vanishing on the diagonal $Y = X$, which Lemma 5 cannot see.

4. Step 3: locality on ℓ^∞

For $x \in \ell^\infty$ write $x|_{\geq N}$ for x with its coordinates below N set to 0.

Proposition 9. *Under* (K3)*, a bounded linear functional φ on ℓ^∞ with $\varphi(e_i) = 0$ for all i is zero.*

Proof. φ kills finitely supported sequences, so $|\varphi(x)| \leq \|\varphi\| \sup_{i \geq M} |x_i|$ for every M. Suppose $\varphi \neq 0$. For every $0 < s < \|\varphi\|$ there is w with $\varphi(w) = 1$ and $\|w\| \leq 1/s$; fix one such vector w_0. Because $r_{n+1} < r_n$ we can choose $s_n \in \ell^\infty$ with $\varphi(s_n) = c_{n+2} - c_{n+1}$ and $\|s_n\| \leq r_n/\|\varphi\|$: take $s_n = (c_{n+2} - c_{n+1})w$ with $\|w\| \leq r_n/(r_{n+1}\|\varphi\|)$. Let $x_n := c_1 w_0 + \sum_{k<n} s_k$; then $\varphi(x_n) = c_{n+1}$ and $\|x_{n+d} - x_n\| \leq r_n/\|\varphi\|$ by the ultrametric inequality. The diagonal sequence $y_i := x_i(i)$ is bounded, since $\|x_i\| \leq \max(\|x_0\|, r_0/\|\varphi\|)$, and $|y_i - x_n(i)| \leq r_n/\|\varphi\|$ for $i \geq n$. Hence $|\varphi(y) - c_{n+1}| = |\varphi(y - x_n)| \leq r_n$, and since $|c_{n+1} - c_n| \leq r_n$ the ultrametric inequality gives $|\varphi(y) - c_n| \leq r_n$ for every n. So $\varphi(y) \in \bigcap_n B(c_n, r_n)$, a contradiction. □

This is the direction of van der Put's theorem [2] that we need, $(\ell^\infty/c_0)' = 0$; the index shift absorbs the usual $(1 + \varepsilon)$ loss, and no Hahn–Banach argument is used.

Proposition 10 (multilinear Phillips property)**.** *Let λ be a bounded d-linear form on ℓ^∞ and $\varepsilon > 0$. There is N such that $|\lambda(u_1, \ldots, u_d)| \leq \varepsilon$ whenever $\|u_j\| \leq 1$ for all j and $u_j = u_j|_{\geq N}$ for some j.*

Sketch (the complete proof is in the Lean development)*. The case $d = 1$, $\lambda = \varphi$.* (a) $\varphi(e_i) \to 0$. Otherwise $|\varphi(e_{i_k})| \geq \varepsilon$ along a subsequence. Let T be the bounded operator with $T(x)_{i_k} = x_k/\varphi(e_{i_k})$ and all other coordinates 0, and $\psi := \varphi \circ T$, so that $\psi(e_k) = 1$ for all k. With sh the right shift, $\psi \circ (1 - \mathrm{sh})$ kills every e_k, hence vanishes by Proposition 9; but $(1 - \mathrm{sh})\mathbf{1} = e_0$, so $0 = \psi(e_0) = 1$. (b) For fixed x, $\varphi(x|_{\geq N}) - \varphi(x|_{\geq N+1}) = x_N \varphi(e_N) \to 0$, so $\varphi(x|_{\geq N})$ converges; the limit is a bounded linear functional of x killing every e_i, hence it is 0. (c) If the claim failed for ε, then (b) would give unit vectors w_k with disjoint finite supports moving to infinity and $|\varphi(w_k)| > \varepsilon$; the functional $c \mapsto \varphi(\sum_k c_k w_k)$ (coordinatewise sum) then contradicts (a).

The case $d > 1$, by induction on d. Suppose the claim fails for ε. First, for every M there is a unit tuple with $|\lambda| > \varepsilon$ *all* of whose slots vanish below M: start from a witness one of whose slots is supported very far out, and split any other slot into its part below M, a finite combination of the e_i, and its tail; the inductive hypothesis, applied to the finitely many $(d-1)$-linear forms obtained by freezing that slot at e_i, $i < M$, shows that the part below M contributes at most ε, so the tail carries $|\lambda| > \varepsilon$. Next, by (b) each slot can be cut down to a finite window. Finally choose such tuples $z^{(s)}$ with windows moving to infinity fast enough and let x be the coordinatewise sum of their first slots. Then $\lambda(x, z_2^{(s)}, \ldots, z_d^{(s)})$ equals $\lambda(z^{(s)})$, plus finitely many terms with an earlier first slot (each $\leq \varepsilon$ by the inductive hypothesis for $\lambda(z_1^{(r)}, \cdot)$, $r < s$), plus one term with a far first slot ($\leq \varepsilon$ by the case $d = 1$). So it has absolute value $> \varepsilon$ for every s, contradicting the inductive hypothesis for the $(d-1)$-linear form $\lambda(x, \cdot)$. □

This is exactly where ℓ^∞ and (K3) are used; the statement is false for c_0 and for spherically complete fields.

5. Step 4: gluing

Choose inductively blocks $I_1 < I_2 < \cdots$ with $|I_t| = 2t$ and vectors $a_t, b_t \in k^{I_t}$ in the unit polydisc as follows. Given the earlier stages let $a' = \sum_{s<t} a_s$, $b' = \sum_{s<t} b_s$, and apply Corollary 8 to get a_t, b_t with $|\det G_t| \geq 1$, where $G_t = \Phi(a' + a_t, b' + b_t; a' + a_t, b' + b_t)|_{I_t \times I_t}$. Put $\varepsilon_t := \frac{1}{2} C^{-2t}$ and use Proposition 10 (with $d = 4$) for the finitely many entry forms $\Phi(\cdot)_{pq}$,

$p, q \in I_t$, to place all later blocks so far out that these forms are at most ε_t on unit arguments one of which is supported there.

Let $a, b \in \ell^\infty$ be defined by $a|_{I_t} = a_t$, $b|_{I_t} = b_t$ and zero elsewhere (these are not norm-convergent sums), and $\mathcal{O} := \Phi(a, b; a, b) \in F$. Splitting $a = a^{\le t} + a^{>t}$, $b = b^{\le t} + b^{>t}$ and using multilinearity, $\|\mathcal{O}|_{I_t \times I_t} - G_t\| \le \varepsilon_t$. All entries are bounded by C. Expanding the determinant multilinearly in columns, every term containing a column of the perturbation is at most $\varepsilon_t C^{2t-1} < 1$, so the ultrametric inequality gives

$$|\det \mathcal{O}|_{I_t \times I_t}| = |\det G_t| \ge 1 \qquad \text{for every } t.$$

On the other hand $\mathcal{O} \in F$ is within $\eta := (2C)^{-1}$ of a matrix R of some finite rank r. Expanding $\det(R + (\mathcal{O} - R))$ on $I_t \times I_t$ in the same way, terms with more than r columns from R vanish, so $|\det \mathcal{O}|_{I_t \times I_t}| \le C^r \eta^{2t-r} < 1$ once $t > r$. This contradiction proves Theorem 1. □

6. An explicit field

Let k_0 be an infinite field of characteristic 2 and $\mathcal{P} = \bigcup_j k_0((t^{1/2^j}))$ the field of 2-power Puiseux series with $|f| = 2^{-\operatorname{ord} f}$; let k be its completion. Then (K1) holds, and (K2) holds because the nonzero constants and their differences have absolute value 1. For (K3) let $\gamma_m = 1 - 2^{-(m+1)}$, $c_n = \sum_{m \le n} t^{\gamma_m}$ and $r_n = 2^{-\gamma_{n+1}}$: then $|c_{n+1} - c_n| = r_n$, the radii decrease strictly, and $r_n > \frac{1}{2}$. If some $a \in k$ lay in all the balls, pick $y \in \mathcal{P}$ with $|a - y| < \frac{1}{2}$; then y lies in all of them too. But y has exponents in $2^{-j}\mathbb{Z}$ for some j, while γ_j has denominator 2^{j+1}, so for $n \ge j$ the series $y - c_n$ has a nonzero coefficient at t^{γ_j} and $|y - c_n| \ge 2^{-\gamma_j} > r_n$. (The completion of an algebraic closure of $\mathbb{F}_2((t))$ is another example.)

7. Remarks

Remark 11 (Why characteristic 2, why ℓ^∞, why this F). In characteristic $\neq 2$ the lift $\frac{1}{2}\big(m \circ (f, g) + m \circ (g, f)\big)$ exists, and the parity argument breaks (pairs add instead of cancelling). If E has an orthogonal base (e_i), e.g. $E = c_0$, and F is Banach, then

$$P(f, g)(m)(x, y) = \sum_{i<j} (x_i y_j - x_j y_i)\, m(f e_i, g e_j)$$

is a bounded lift in every characteristic, so Q is analytic; a similar formula works for a base of E', and t-orthogonal bases suffice. This covers finite-dimensional E and all Banach spaces over a discretely valued k [3], so the counterexample needs a dense valuation; one can also show that Q is analytic when F is spherically complete. On c_0 the glued vectors a, b of Step 4 do not exist and Proposition 10 fails. For $E = \ell^\infty$ and the larger target $\ell^\infty(\mathbb{N} \times \mathbb{N}, k) \supseteq F$ a bounded lift also exists (sort the indices entrywise), but it sends W to matrices such as the "staircase" $\mathbf{1}_{\{(p,q)\,:\,p \text{ odd}, q \text{ even}, p<q\}}$, which are not limits of finite-rank matrices, so it leaves F. These positive statements are not proved here and are not part of the formalization.

Remark 12 (Higher degree, other characteristics). On paper, in characteristic 2 every $|\iota| \ge 2$ is covered by wedging with a coordinate of $E \oplus k$; this is not formalized. In odd characteristic ℓ, for forms of degree ℓ, the question is open: odd-degree forms have no Pfaffian-type top power, so a different invariant is needed.

8. Formalization

The Lean 4/Mathlib development is at

https://github.com/Deicyde/alternating-analytic-counterexample

The library `AltCounter/` contains no `sorry`, and the only axioms it uses are the three standard ones (`propext`, `Quot.sound` and `Classical.choice`). Its headline theorem assumes that Mathlib's

`ContinuousAlternatingMap.compContinuousLinearMapCLM`

is C^ω for all nontrivially normed fields, normed spaces and finite index types, and derives `False`; a second theorem does the same from the weaker statement in Theorem 2. Both are restated in a Mathlib-only file `Challenge.lean` (with `sorry` in place of the proofs, as that format requires) and checked against `Solution.lean` by the `leanprover/comparator` tool. Steps 1–4 and §6 are formalized, including the complete proof of Proposition 10 by induction on d; the Lean constants in Step 4 are chosen with more slack than here. Not formalized: non-analyticity at base points other than 0, index types with more than two elements, the positive statements in the remarks, and the bundle-level consequence (no C^ω vector-bundle structure on bundles of continuous alternating maps in this generality).

References

[1] J. McCarthy, *Smoothness reflects along a linear isometry with closed range*, draft pull request #43548 to Mathlib, https://github.com/leanprover-community/mathlib4/pull/43548.
[2] M. van der Put, *Espaces de Banach non archimédiens*, Bull. Soc. Math. France **97** (1969), 309–320.
[3] A. C. M. van Rooij, *Non-Archimedean Functional Analysis*, Marcel Dekker, 1978.